

SAKET BISHNU

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Professional Summary

Computer Science student specializing in Big Data Analytics with strong interest in software Development. Skilled in programming, Machine Learning, and Deep Learning with hands-on project experience. Passionate about solving real-world problems using data-driven solutions and AI technologies.

Education

B.Tech – Computer Science & Engineering (Big Data Analytics)

2022 – 2026

SRM Institute of Science and Technology, Kattankulathur

CGPA: 8.2 / 10

Skills

Languages & Frameworks: Python (Core OOP), C++, SQL, NumPy, Pandas, Scikit-learn, TensorFlow, Excel

Tools & Technologies: GitHub, Anaconda, Power BI, Docker, Jupyter, SDLC, Debugging, Web Scraping

ML / Data Skills: Data Preprocessing & Cleaning, Feature Engineering, Model Training & Evaluation

Areas of Interest: Machine Learning, Computer Vision, Automation Testing, Model Deployment, Data Science

Projects

LiverSegNet – Surgical Vision AI (Medical Image Segmentation)

Apr 2026 – Present

- Built a DeepLabV3+ (ResNet-50) semantic segmentation model achieving **88% Dice score** on CholecSeg8k, simultaneously segmenting liver anatomy and surgical instruments from laparoscopic video frames.
- Engineered a robust data pipeline with augmentation (flip, rotate, elastic distort), mask normalization, and weighted Dice loss to handle smoke, glare, occlusion, and motion artifacts in **40000 annotated frames**.
- Implemented a PyTorch training pipeline with Dice + Cross-Entropy loss, data augmentation, and evaluation metrics (Dice, IoU, pixel accuracy) to improve segmentation performance.

GenAI Video Knowledge Retrieval System (LLM + RAG Pipeline)

Oct 2025 – Dec 2025

- Built a RAG pipeline that converts educational videos into searchable knowledge bases using FFmpeg and OpenAI Whisper, processing **10+ hours** of lecture content.
- Implemented semantic retrieval with BGE-M3 embeddings and cosine similarity, achieving **top-3 answer accuracy of 91%** on a 500-query test set.
- Integrated DeepSeek/Llama 3 via Ollama to generate timestamp-specific, context-aware answers with response latency on local hardware.

Vendor Performance Monitoring System (SQL | Python | Power BI)

Sep 2025 – Nov 2025

- Designed and optimized advanced SQL queries using CTEs, joins, and aggregations to analyze vendor performance metrics across **12M+** sales records and **2M+** purchase records stored in SQLite.
- Built an automated ETL pipeline in Python using Pandas, SQLAlchemy, and SQLite to ingest, clean, transform, and merge multi-source inventory, sales, purchase, and freight datasets into a consolidated **3 hours/week**.
- Developed interactive Power BI dashboards for vendor profitability, sales-to-purchase ratio, freight cost analysis, and stock turnover, enabling data-driven insights across multiple vendors and product categories.

Extracurricular Activities

Social Work Teaching Intern

june 2024 – july 2024

Social Warriors Foundation

- Delivered **30+ academic and life-skills sessions** to underprivileged students, improving average class attendance by **18%**.
- Led a team of **15 volunteers** to organise fundraising events, raising resources for learning materials benefitting **100+ students**.
- Developed strong leadership, communication, and organizational skills while guiding and coordinating with peers. Boosted participation and engagement by introducing creative and innovative event idea.

Achievements & Certifications

Achievements

- 1st Prize – TechKnow 2022-2023 (Semiconductor Expo, SRMIST)
- Secured 4th Position in the TECH-Q Quiz Contest
- Participant – OODP HACK'23 (SRMIST Hackathon)
- Competitive Programming & Cracking MAANG Workshop at SRMIST

Certifications

- Project Management – NPTEL
- Analytics with SQL and Python
- Data Science Course – CodeWithHarry
- Student Member at ISTE